

A Robust Vision-Based Dynamic Sign Language Recognition Using a Hybrid CNN-LSTM Model

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Abstract—To recognize meaningful movements of hand gestures, Sign language recognition is important. It is a significant solution in intelligent communication between the deaf community and hearing societies. The general population's lack of proficiency in sign language impedes meaningful interactions with individuals who depend on it. Although many studies have been conducted on this topic, the development of static and dynamic sign language recognition models continues to hold significant challenges in the field of research. This paper presents a novel framework for implementing a vision based Sinhala Sign Language recognition model using deep learning approaches. Our proposed methodology includes image acquisition, pre-processing, detection, and classification stages for a complete recognition. In the classification stage, the CNN-LSTM model achieves 94% accuracy for around 20 signs. This method identifies and interprets unique common signs and gives a meaningful sentence in text. The proposed approach shows promising results for interpreting both static and dynamic signs of Sinhala sign language.

Index Terms—CNN, Deep Learning, Dynamic Sign Language Recognition, LSTM, Sign Language

I. INTRODUCTION

Nearly 70 million people worldwide experience disabling hearing loss, according to current statistics. A deaf person can communicate thoughts using a specific hand gesture and facial expressions rather than words. Sign language gestures are accurately understood by only a limited group of people, such as instructors and family members of individuals who are deaf. Therefore, current translation services are either human-based or expensive. Sign Language Recognition (SLR) is an effective method that reduces the barrier between deaf communities and hearing societies [1].

It tends to be a widely discussed topic with the increasing research interests and interaction between humans and machines. Using advanced technology, this problem can overcome the communication difficulties they face.

Sign language uses multiple body parts, including fingers, hands, arms, head, body, and facial expressions, to communicate information [2]. However, our proposed methodology only focuses on recognizing sign language through hand gestures. Sign languages are complex linguistic systems, not merely simple gestures. They break down into smaller components like hand configuration, location, and movement. These elements work together to form the language's grammar and vocabulary, much like how sounds and words function in spoken languages [3]. There are two forms of hand gestures: static and dynamic. Static gestures include performing only hand poses, whereas dynamic gestures include the movements of the hand to represent sign language words [4].

Consequently, recognizing dynamic signs poses challenges due to their complexity and extensive variations in vocabulary, potentially leading to misinterpretations of significant differences among signers. Therefore, focusing on dynamic SLR presents a more effective approach to addressing these issues. However, dynamic SLR faces challenges in detecting the hand location while moving and identifying the shape and direction [5]. Recognition becomes more complex due to the gestures performed by different people, which differ in spatial and temporal variation. Furthermore, selecting an appropriate classifier is critical for achieving accurate recognition results. Therefore, we have used effective pre-processing, detection, and classification methods to overcome the challenges.

Hand gesture recognition can be achieved through either a

vision-based or sensor-based approach. Vision-based methods capture images or video of hand gestures using a camera, and sensor-based methods employ a sensor instrument (Glove) to track and record the motion, position, and velocity of the hand. As deep learning continued to evolve, researchers directed their attention toward vision-based SLR methods, employing cameras to capture data and streamline the process for a more intuitive experience. The primary steps of a vision based recognition system are frame acquisition, image pre-processing, detection, and classification.

The sequel of the paper is organized as follows. Related works are described in Section 2. Section 3 provides the proposed methodology describing the major stages: dataset, pre-processing & detection, and classification. The results and discussion are included in Section 4. Finally, Section 5 summarizes the conclusion and future work.

II. RELATED WORKS

In this section, prior research approaches for sign language translation are discussed.

The sign language gestures are input into the recognition system as images. These images are either captured by frames from videos or directly by devices such as web cameras [5], [6], Kinect cameras [7], or CMOS image sensor cameras [8].

In [9], authors perform a binary mask through morphological transforms to isolate the critical regions and the median blue filtering method has been utilized for edge detection. Similarly, [8] and [10] utilize morphological filtering. While [8] uses the Otsu method for hand segmentation, [10] uses a customized threshold method. Both [11] and [12], first perform face detection and face elimination to ease the process of skin segmentation.

Sign gestures are detected from the pre-processed images in [6] and [13] using the MediaPipe (MP) approach for real-time detection. It allows the detect both hand and body movements. Both [9] and [12] propose hand detection and tracking algorithms where the larger contours were extracted eliminating further noise in the pre-processed image. In [10], authors employ an algorithm where they input the segmented image and output three features of the hand - height, width, and centroid. Similarly, [8] utilizes an algorithm that extracts four features - area of the hand, perimeter of the hand, detection of the thumb, radial profile, and angular position. [5] employs a B3D ResNet model to obtain feature vectors from the pre-processed image sequence. [11] also uses a feature vector of dimension 6, where key frames get extracted based on hand position and the shape changes in the frame sequence. [14] propose a 3D-ResNet module, which employs a sliding window technique to input the selected images from the sequence and output the features. In [15], authors employ a Bayesian approach for skin classification. They used the probabilities of observing skin and not observing skin. Without estimating direct values for the two probabilities they use the Bayes rule to define a threshold value. Hence adaptive skin color detection has been achieved.

The sign language gestures are recognized in spoken language in [9] by utilizing a CNN model (for static signs) and a tree architecture (for the dynamic signs, that are formed by static signs). The images containing the static signs are input to the model and depending on three consecutive recognized signs, 2/3 majority sign is considered the final output. When the classified static signs were ordered in a pre-defined sequence a dynamic sign is recognized. [5] addresses the difficulty of recognizing similar hand gestures apart, with the help of a complex model architecture including convolutional layers, LSTM layers, and a soft-max layer. The proposed method inputs the extracted feature vectors from the video frames and labels each after analysing. Hence the video sequence label will be predicted and then output the prediction with the top score. [6] has used an LSTM Neural Network alone to classify static and dynamic signs. The extracted points from MP Holistic are fed into the model and output the recognized static signs. Aside from the neural network architectures, [7] proposes an algorithm with two steps, step 1 being a Dynamic Time Warping algorithm (DTW) and step 2 being a Nearest Neighbor classification. A dictionary has been implemented with 75 sequences from 15 gestures. DTW calculates each of the two arrays performed by input frames for each sample in the dictionary and outputs the cost function values. Depending on these values, recognition is provided as a word-by-word output from the classifier. In a similar study, [14] employs a model with a soft-DTW constraint along with LSTM and Connectionist Temporal Classification (CTC). [12], utilizes a CNN and LSTM model combination. In this model, the CNN which is composed of max-pooling layers and fully connected layers transforms the pre-processed images into vector representations and inputs them into the LSTM network. Each state of LSTM predicts the label of the sign and helps to recognize the word in the next movement with an accuracy of 98.43%. In [11], the authors propose an SVM architecture with three classification models for Zernike moments-based, trajectory-based, and shape-based recognition.

III. PROPOSED METHODOLOGY

The system mainly consists of the following stages shown in Fig. 1. The proposed system is developed using deep neural network models, building on their proven success in recent gesture recognition research. Specifically, it consists of a hybrid model of CNN and LSTM architecture to extract features and recognize them.

A. Dataset

We created a new dataset with over 2500 sample videos of 22 Sinhala Sign Language (SSL) gestures from 24 people. The respective SSL signs were initially taught by teachers at 'The School for the Deaf, Ratmalana, Sri Lanka'. A USB 2.0 HD UVC WebCam and a Python script leveraging OpenCV were used to collect and store 3-second long videos. We selected words related to interactions between hearing-impaired people and medical centers, grocery stores, and banks. Table 1 illustrates the data set including both static and dynamic signs.

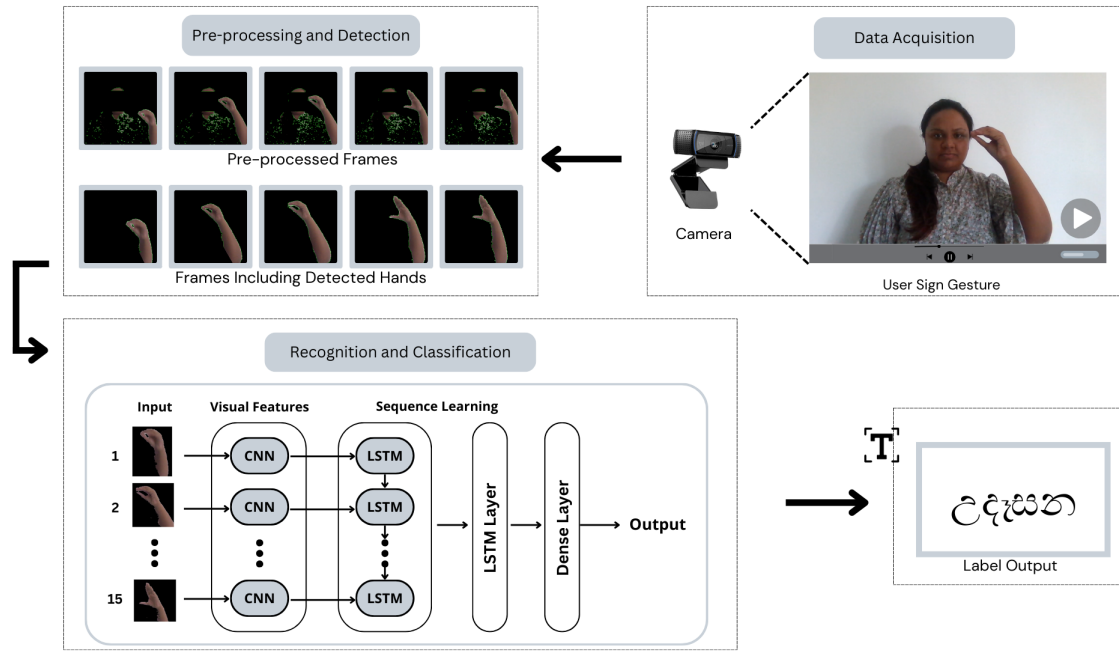


Fig. 1. Overall methodology

TABLE I
COLLECTED GESTURE SIGNS

Sinhalese Word	English Idea	Type
Awe	Came	Dynamic
Epa	Don't	Dynamic
Hadagannai	Remake / Update	Dynamic
Haduna	Felt sick / Had an illness	Dynamic
Hi	Hi	Dynamic
Ithuru	Change (with related to money)	Dynamic
Kauntaraya	Counter (in a bank or medical station)	Dynamic
Kalanthe	Dizziness	Dynamic
Keeyada?/Mudal	How much? / Money	Dynamic
Ko?	Where?	Dynamic
Kohomada?	How are you?	Dynamic
Mata	(to) Me	Static
Nawatha	Again	Dynamic
Oba	You	Static
Pereda	Day before yesterday	Dynamic
Potha	Book	Dynamic
Ridenenawa	Feeling sick / Hurt	Dynamic
Sbha Udasanak	Good Morning	Dynamic
Sthuthiyi	Thank you	Dynamic
Thampath Karana	Store	Dynamic
Thibuna	Had	Static
Ugura	Throat	Dynamic

B. Pre-processing and Detection

This stage is used to improve the overall efficiency and isolate the hand gestures from the background or non-relevant elements of a frame. First to ensure that the face does not interfere with hand gesturing, detecting and masking the face is done using the Haar Cascade Classifier, in the gray scale. After applying masking to the original image, the face is removed from the frame. Then the face-masked image is

converted to YCbCr color space, which is effective for skin color segmentation. To highlight the skin regions from the rest of the background, a binary mask is created corresponding to common skin color tones (20,135,77 & 255,170,138). Then the image is converted to gray scale and noise is reduced using a bilateral filter. Canny edge detection is utilized to highlight the edges within the masked image. To accurately extract the hand contour, contours are subsequently extracted from the detected edges, and the largest contour by area is selected (corresponds to the hand). This contour is used to create a convex hull and a mask is generated accordingly to isolate the hand. Further processing is done to identify the bounding box and the hand is cropped from the rest of the image to feed into the classification model.

C. Classification

Unlike a static sign, a dynamic sign is a combination of static signs distributed over a specific period of time. Because of that, not only the spatial features, but also the temporal features representing the relationship between each individual frame is required to be extracted. For this, our proposed model leverages a deep learning based hybrid architecture consists with a CNN and an LSTM as shown in Fig.2.

A CNN is a multi-layer neural network specifically designed for processing two-dimensional data. By employing concepts such as sparse connectivity and shared weights, CNNs significantly reduce the number of training parameters, enhancing their adaptability to identify patterns in recognizing images. As the first stage of the proposed model, a CNN-based approach is incorporated. Here, the CNN architecture is specifically designed and tuned to leverage the full potential of the hand extracting algorithm which is a novel combination of

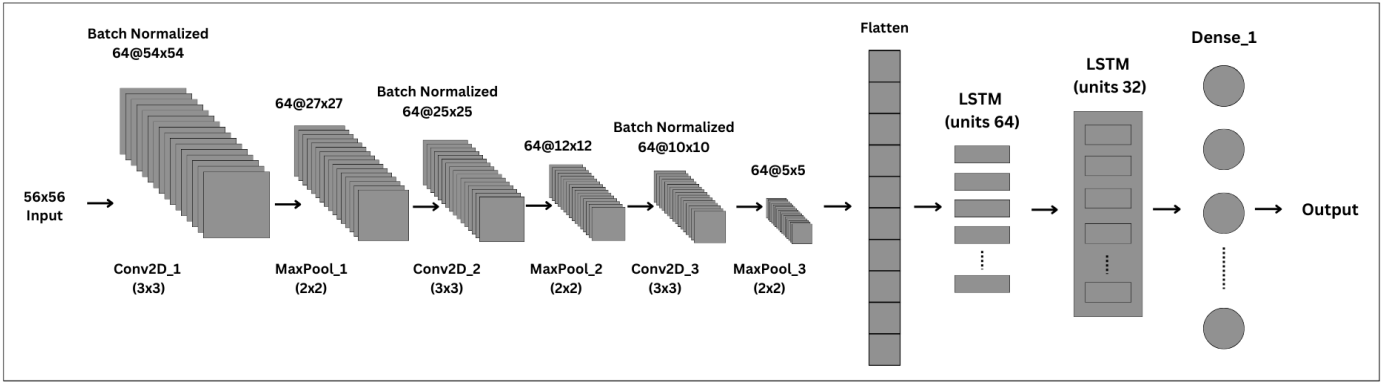


Fig. 2. CNN-LSTM model layer architecture (Considering 1 time step)

already available techniques for the spatial feature extraction of each frame. Basically, a CNN consists of convolutional layers, batch normalization layers and max-pooling layers. The Convolutional layers convolve the input image with its kernels to produce feature maps that capture various patterns. Batch normalization is mainly applied to accelerate the training by normalizing the inputs of each layer where max-pooling layers used to reduce the dimensionality of the input data reducing the computational overhead. The proposed CNN model uses 3 convolutional layers with each having 64 3x3 kernels with a 2x2 max pooling layer for improved feature extraction. This enables the complex feature extraction from less number of pixels with improved computational efficiency. Further, ReLU activation function is used to introduce non-linearity, enabling the network to learn complex and abstract features more effectively.

After extracting the features from a 56x56 pre-processed and hand detected input video frame, the feature maps are flattened into a one-dimensional vector of size 1600, preparing the data for final classification. The flattened vector is then passed to the LSTM model for the next stage of feature extraction, the temporal feature extraction. LSTM is a variation of an RNN model which has the capability of handling long time-series data without getting affected by the issue of vanishing gradient. The proposed LSTM model comprises with two layers: the first layer identifies the short-term temporal patterns and fed to the second layer for recognizing long-term temporal dependencies. By combining the outputs of these two layers, our methodology effectively discerns signs with complex gestures, enhancing classification accuracy and robustness. To avoid network overfitting, a dropout layer is employed with a dropout rate of 0.25 between the two LSTM layers. Finally, the features are fed into a Dense layer for classification. Model is trained using the Adam optimizer with a batch size of 5 and 0.2 validation split.

As the final step, an algorithm based on the prediction probabilities is used to generate the classification results. During the predictions, each continuous frame is assigned a predicted class probability, taking into account both spatial and temporal characteristics. However, this method has some

issues. Directly using these predicted probability values can lead to inaccurate predictions or result in an output class even when no gesture is present in the real-time feed. To resolve this, a threshold based approach is implemented to check whether a fixed number of consecutive frames predict the same class with a higher probability than the threshold. The number of frames and the prediction probability threshold values were determined by analyzing the model's prediction patterns. As a result, this method ensures accurate and continuous sign predictions in a real-time video feed consisting of multiple signs.

IV. RESULTS AND DISCUSSION

In this section, we present and analyze the performance and compare our CNN and LSTM model, with a Transformer architecture and an RNN model. As well as we discuss the novelty of the SSL gesture dataset and the user friendly interface our system uses.

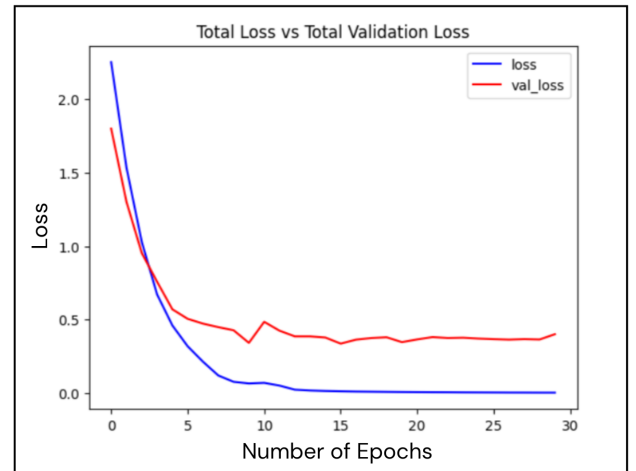


Fig. 3. Training loss graph of CNN-LSTM architecture

As shown in Fig. 3, the validation loss stabilizes around 0.4 after an initial decrease, indicating a reasonable fit with some overfitting towards the later epochs. In the accuracy graph Fig.

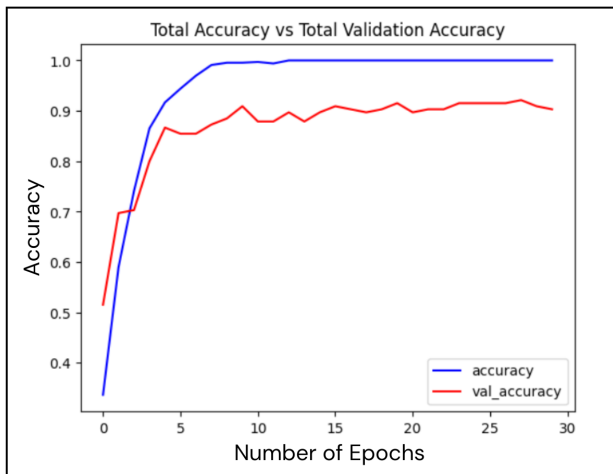


Fig. 4. Training accuracy graph of CNN-LSTM architecture

4, training accuracy reaches 1.0 demonstrating the model's strong learning capacity. Validation accuracy reaches about 0.9 suggesting a good generalization despite the slight overfitting indicated in the loss graph. The overall model achieved a test accuracy of 93.72% with a loss of 0.2921. Achieved accuracies for a set of sign gestures are shown in Fig. 6. Two benchmark models were developed and fine-tuned as hybrid architectures, one being CNN and Transformer and another one as CNN and RNN models. Accuracies are 85.09% and 86.91%, respectively as illustrated in Fig. 7 and Table 2. While the CNN model analyses the sequential data, the Transformer and RNN models analyze the temporal data in the extracted features.

The ability of LSTM to retain information with memory cells for longer periods allowing to capture long-term dependencies of complex dynamic signs outperforms the traditional RNN which struggles with issues like vanishing gradients. When considering the transformer, LSTM has performed well due to its stability in training for sequential data with a comparatively smaller dataset. With transformers, having a larger amount of data is important to achieve the best performance which may not always be available with sign language datasets. In terms of accuracy, scalability and higher generalization CNN-LSTM model architecture shows a robust performance.

The proposed CNN-LSTM architecture leverages the full potential of the proposed hand detection algorithm enabling the learning and classification of complex hand gestures. The model performs well with complex and lengthy dynamic signs with the used CNN architecture for spatial feature extraction and 2 layered LSTM architecture for temporal feature extraction. Also, the model performs well in identifying 2 signs separately which are almost similar statically but different dynamically.

The paper presents a new dataset including 22 SSL gestures that could be used in day to day scenarios. There are some studies carried out with invented datasets for SSL but most of them are static signs [7], [9]. But our dataset (over 2500

sample videos) mostly contains dynamic signs which are more practical gestures used in SSL. In [11], authors implement an Indian Sign Language dataset including alphabet (static signs) and dynamic signs (videos) from 7 people. While collecting the images and videos, wearing at least half sleeves is a must. However implementing our dataset does not require any special clothing. Using 24 SSL users enriched the dataset with diverse and comprehensive data.

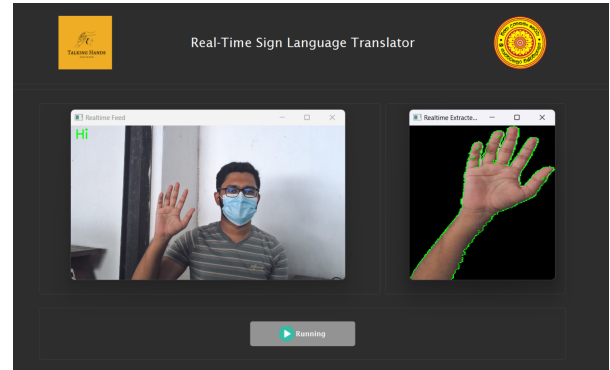


Fig. 5. User interface

Our system was developed as a full-duplex communication system to facilitate meaningful conversations through audio dialogs in real time. The system translates sign language into text and audio for the normal hearing people and converts spoken audio into readable text, enabling hearing-aided individuals to follow the conversation seamlessly. As the end result, we designed a user-friendly application with an intuitive interface. This user interface works as the front-end for the backend process of capturing video frames, processes them in real-time, displays the translated output, and generates audio responses, enabling seamless and effective two-way communication.

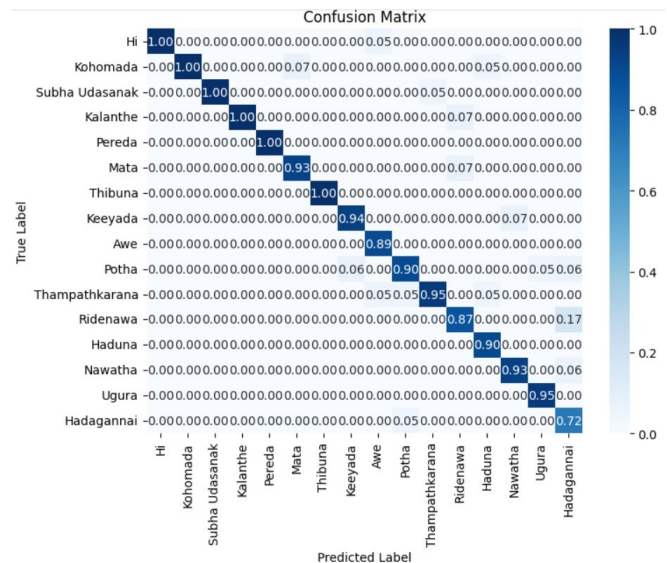


Fig. 6. Accuracies for each sign gesture

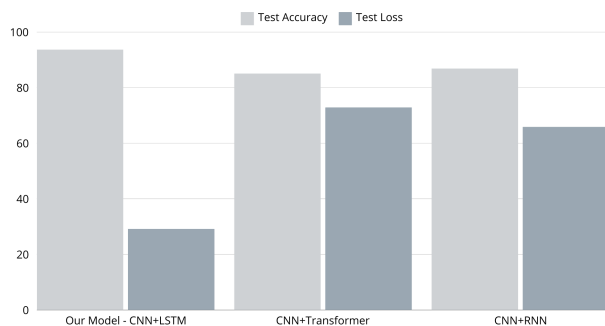


Fig. 7. Testing accuracy and testing loss comparison

TABLE II
COMPARISON OF CNN-LSTM, CNN-TRANSFORMER, AND CNN-RNN
MODEL PERFORMANCE

Metric	CNN-LSTM Our Model	CNN- Transformer	CNN-RNN
Training Accuracy	99.89%	96.34%	99.59%
Training Loss	0.2921	0.1423	0.0161
Test Accuracy	93.72%	85.09%	86.91%
Test Loss	0.2921	0.7294	0.6592

V. CONCLUSION AND FUTURE WORK

This paper presents a novel framework for recognizing continuous SSL using a hybrid CNN-LSTM architecture. Our model effectively handles both static and dynamic signs with a test accuracy of 93.72% with a loss of 0.2921 outperforming the baseline models. This research contributes to the field of SSL recognition with the data set (more than 2500 sample videos) for future researchers and offers better communication between the deaf community and hearing individuals. Since we use constant threshold values for skin color segmentation, the hand may not be accurately detected in poor lighting conditions. Adaptive thresholds and the lighting compensation using CLA Histogram Equalizer can be used to overcome this issue. Further morphological operations can be used to refine hand boundaries.

To ensure practical applicability, we tested our system with the hearing aid users and achieved promising results for the selected set of signs. This demonstrates the effectiveness of the system while also highlighting areas for improvement. We believe the accuracy can be further enhanced by expanding the dataset to include a broader range of gestures and optimizing the model for real-world situations. Future work will focus on these improvements to make the system more robust and versatile for practical use in diverse communication scenarios.

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